How-To Library

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Importing and Exporting Data Assets Between Microsoft SQL Server and Enterprise Data Preparation

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Abstract

You can import data from Microsoft SQL Server database tables into your data lake. You can also export data from your data lake to a Microsoft SQL Server database. Review this article to prepare Enterprise Data Preparation to import and export data assets.

Supported Versions

- Enterprise Data Lake 10.2 x
- Enterprise Data Lake 10.4.0

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Overview

You can import data from Microsoft SQL Server database tables into your data lake. You can also export data from your data lake to a Microsoft SQL Server database.

After you import a data asset from SQL Server, you can add the imported data to a project in Enterprise Data Preparation as a worksheet. After you prepare an asset and export the data to SQL Server, you can perform further operations on the data in SQL Server.

To prepare to import and export data assets between SQL Server and Enterprise Data Preparation, complete the following steps:

1. Download the Microsoft JDBC Driver for SQL Server, and then copy the driver to the Data Integration Service machine associated with Enterprise Data Preparation.
2. Create one or more JDBC connections to import and export data assets. You might create different connections for import and export operations.
3. Add the DoNotUseOwnerNameForSqoop custom property to the Enterprise Data Preparation Service.

Download the JDBC Driver

Download the Microsoft JDBC Driver for SQL Server, and then copy the driver to the Data Integration Service machine associated with Enterprise Data Preparation.

When you import or export data, the Data Integration Service uses Sqoop to connect to the cluster, and then it runs a Sqoop job on the cluster to import or export the data. Sqoop uses the driver to connect to the SQL Server database target when the import or export job runs on the cluster.

Download the Microsoft JDBC Driver for SQL Server .jar file that corresponds to your preferred Java Runtime Environment (JRE) settings. You can download the .jar file from the following link:

https://www.microsoft.com/en-us/download/search.aspx?q=jdbc

Copy the .jar file to the following directory on the machine where the Data Integration Service runs:

```
<Informatica installation directory>/externaljdbcjars
```

Create a JDBC Connection

Create one or more JDBC connections to import and export data assets. You might create different connections for import and export operations.

You include the arguments Sqoop uses to connect to the SQL Server database in each JDBC connection.

1. In the Administrator tool, click the **Manage** tab > **Connections** view.
2. In the Domain Navigator, click **Actions > New > Connection**.

   The New Connection dialog box appears.
3. In the New Connection dialog box, select **Database** > **JDBC**.
4. Click **OK**.

   The **New Connection** wizard appears.
5. Enter the JDBC connection properties required to connect to the SQL Server database.

   The following table describes the connection properties you enter:

Property	Description
Name	Name of the connection. The name is not case sensitive, and must be unique within the domain. The name cannot exceed 128 characters, contain spaces, or contain the following special characters: `~ ` ! $ % ^ & * ( ) - + = { [}] | \ : ; " ' < , > . ? /`
ID	String that the Informatica services use to identify the connection. The string is not case sensitive. The string must be 255 characters or less, and must be unique in the domain. Default value is the connection name.
Description	Description of the connection. The description cannot exceed 765 characters.
User Name	Database user name.
Password	Password for the database user name.
JDBC Driver Class Name	Name of the JDBC driver class packaged with Informatica. Enterprise Data Preparation uses the driver to connect to the SQL Server database. Enter the following value: `com.informatica.jdbc.sqlserver.SQLServerDriver`

Connection String	Connection string to connect to the target. Use the following format for the connection string:
	`jdbc:informatica:sqlserver://<database host name>:<database port>;databaseName=<database name>; charset=UTF16`
	The following example shows a sample connection string:
	`jdbc:informatica:sqlserver://sqldb01.company.com:1433;databaseName=edlsqldb;charset=UTF16`

Environment SQL	Optional. SQL commands to set the database environment when you connect to the target. The Data Integration Service executes the connection environment SQL each time it connects to the target.
Transaction SQL	Optional. SQL commands to set the database environment when you connect to the target. The Data Integration Service executes the SQL at the beginning of each transaction.

Property	Description
SQL Identifier Character	Select " " from the drop down list.
Support Mixed-case Identifiers	Select **Yes** from the drop down list.
Use Sqoop Connector	Select **Sqoop v1.x** to enable Sqoop connectivity.

Sqoop Arguments	Sqoop arguments required to connect to the target.
	Specify the connection string Sqoop uses to connect to the target in the `--connect` argument. Specify the class name of the Microsoft JDBC Driver for SQL Server driver Sqoop uses in the `--driver` argument.
	Use the following format for the argument string:
	`--connect jdbc:sqlserver://jdbc:informatica:sqlserver://<database host name>:<database port>;databaseName=<database name>;charset=UTF16 --driver <JDBC driver name>`
	The following example shows a sample argument string:
	`--connect jdbc:sqlserver://sqldb01.company.com:1433;database=edlsqldb;charset=UTF16 --driver com.microsoft.sqlserver.jdbc.SQLServerDriver`

6. When you finish entering the connection properties, click **Test Connection** to test the connection.
7. Click **Finish**.

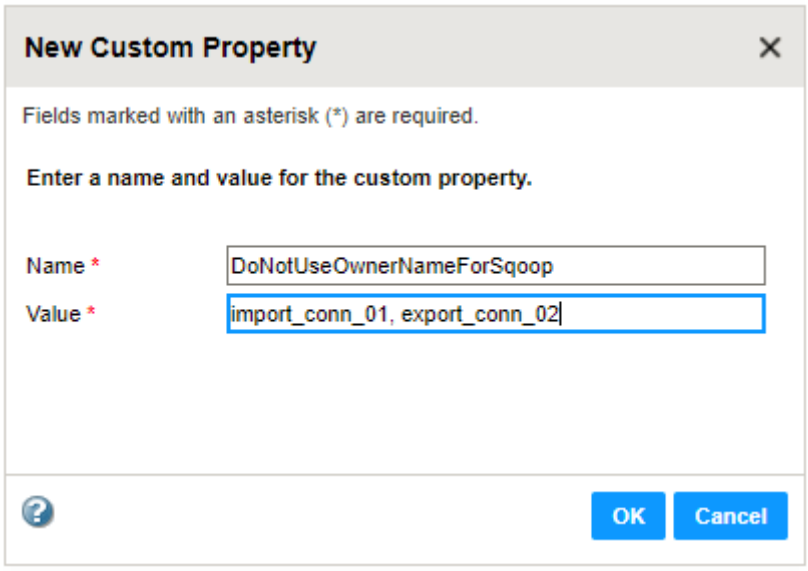
Configure the DoNotUseOwnerNameForSqoop Custom Option

Use the Administrator tool to add the DoNotUseOwnerNameForSqoop custom property to the Enterprise Data Preparation Service. Enter the name of each JDBC connection that Enterprise Data Preparation uses to import and export data as the custom property value.

The DoNotUseOwnerNameForSqoop custom property enables Sqoop to use the connections to access a table in the SQL Server database target without requiring the name of the schema that contains the table. By default, Sqoop requires the name of the schema that contains the table to connect.

1. In the Administrator tool, click the **Manage** tab > **Services and Nodes** view.
2. In the Domain Navigator, select the Enterprise Data Preparation Service.
3. Click the **Properties** view.
4. Click the pencil icon in the Custom Options section.
5. Click **New** in the Edit Custom Options dialog box.
6. Enter DoNotUseOwnerNameForSqoop as the custom property name.
7. Enter the name of each JDBC connection Enterprise Data Preparation uses to import and export data as the custom property value. If you enter multiple connection names, separate each connection name with a comma.

The following image shows the **New Custom Property** dialog box:



8. Click **OK**.

You can now import and export data assets between Enterprise Data Preparation and SQL Server.

For more information about importing and exporting data assets, see the "Import, Export, and Upload Data" chapter in the Enterprise Data Preparation User Guide.

Authors

Ram Singh

Aravind Kumar Arunachalam

Swathi N.R.

Dan Hynes